

# CLINICAL PRACTICE UPDATES

## AGA Clinical Practice Update on Endoscopic Lifting Agents: Commentary



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**DESCRIPTION:** This American Gastroenterological Association (AGA) Institute Clinical Practice Update (CPU) reviews endoscopic lifting agents and their use in the gastrointestinal tract.

**METHODS:** This CPU was commissioned and approved by the AGA Institute Clinical Practice Updates Committee and the AGA Governing Board to provide timely guidance on a topic of high clinical importance to the AGA membership. This CPU underwent internal peer review by the Clinical Practice Updates Committee and external peer review through standard procedures of *Clinical Gastroenterology and Hepatology*. This expert commentary incorporates important as well as recently published data in this field and reflects the experiences of the authors.

**Keywords:** Colorectal Polyps; Endoscopic Mucosal Resection; Endoscopic Submucosal Dissection; Peroral Endoscopic Myotomy; Submucosal Injection.

### Introduction

Gastroenterologists frequently encounter and resect pre-cancerous lesions and early cancers of the gastrointestinal tract. Endoscopic resection has evolved from snare polypectomy to endoscopic mucosal resection (EMR), and now, endoscopic submucosal dissection (ESD).<sup>1,2</sup> For decades, “homemade” solutions were used to lift lesions prior to removal. The sentinel lifting agent was normal saline, which later was mixed with a blue contrast agent, usually indigo carmine or methylene blue. The primary benefit of submucosal lifting is the creation of a cushion between the lesion and muscularis propria (MP). This submucosal cushion separates the lesion from the MP, reducing the risk of immediate or delayed perforation. The addition of a contrast agent demarcates lesion margins and stains the submucosa, which is fundamental to ESD and allows for assessment of MP injury during EMR.<sup>3</sup> The tendency of saline to dissipate quickly prompted investigations into viscous solutions, which dissipate more slowly. Several of these solutions were commonly used prior to United States Food and Drug Administration (FDA) approval of any agent. Many of the active ingredients in FDA-approved solutions were tested in investigator-initiated studies prior to their commercial development. In 2015, compound SIC-8000 became the first FDA-approved submucosal injection agent. Since then, several other injection fluids have been brought to market, although “homemade” agents remain available for use.

This Clinical Practice Update reviews available agents and provides a clinically relevant commentary on their indications and use. This article is not a formal systematic review but is instead focused on practical advice for endoscopists. No formal rating of the quality of evidence or strength of recommendations was performed.

### When to Inject

Common questions during endoscopic resection include deciding whether to and which solution to inject. ESD requires submucosal injection, regardless of anatomic location. Some endoscopists performing ESD start the submucosal injection and incision using a pre-packaged viscous solution. The endoscopist may continue with the viscous fluid, or transition to saline or another less expensive solution. A pump or water-jet facilitates submucosal injection and speeds dissection. For peroral endoscopic myotomy (POEM), normal saline with a blue contrast agent is instilled into the submucosa with water.

For EMR, the fluid selected is determined by lesion size, predicted histology, and endoscopist preference. Based on the United States Multi-Society Task Force



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(USMSTF) on Colorectal Cancer, submucosal injection is optional for non-pedunculated colorectal lesions (NPCRLs) of intermediate size (10–19 mm).<sup>4</sup> A subsequent study found that cold snare polypectomy without submucosal injection was non-inferior to other resection methods that utilized a submucosal injection, for the resection of NPCRLs  $\leq 15$  mm.<sup>5</sup> The USMSTF considers EMR first-line therapy for most NPCRLs  $\geq 20$  mm and advocates viscous solutions (VSs) as preferred. VSs allow resection in less time and in fewer pieces.<sup>4</sup> The benefits of viscous solutions pertain particularly to large non-pedunculated adenomas. Sessile serrated lesions (SSLs) have minimal submucosal fibrosis and lift adequately with any solution, including saline. Thus, SSLs have proven ideal for cold EMR (ie, EMR without electrocautery).<sup>6</sup> SSLs do not always require injection, especially those  $< 20$  mm in size or when the margins are clearly visible without injection.<sup>7</sup> The use of lifting agents for pedunculated polyps is generally at the discretion of the endoscopist.

For cold EMR, the addition of dilute epinephrine (usually 1:200,000 to 1:500,000) provides a cleaner working field with less intraprocedural oozing. Epinephrine is also often used for EMR with electrocautery.<sup>8</sup> Serial injections of dilute epinephrine into the submucosa may cause post-procedure abdominal pain, resulting in diagnostic confusion between perforation, post-polypectomy syndrome, and expected post-procedural pain in the first hour post-procedure.<sup>9</sup>

An alternative to conventional (inject and resect) EMR is underwater EMR (U-EMR), which is performed without submucosal injection.<sup>10</sup> Similarly, papillectomy does not generally require submucosal injection, but normal saline (NS) with contrast may provide reassurance there is no muscle injury to the duodenal wall, especially when papillary lesions involve the surrounding mucosa.

## How to Inject

A retractable needle catheter is first “primed” to avoid air injection.<sup>11</sup> The ease of locating the submucosal space varies with location; thus, access is typically easier in the small bowel than the colorectum. Submucosal fibrosis, including that induced by previous partial resection, biopsy, or tattoo, can make access difficult and create an iatrogenic non-lifting sign.<sup>12</sup> Thinner needles (eg, 25-gauge vs 23-gauge) may facilitate access. When injecting inexpensive agents such as saline or hydroxyethyl starch, the submucosa can be accessed by placing the needle tip on the desired point of injection, beginning fluid injection, and then advancing the needle into the tissue while injection proceeds. The fluid emanating from the needle expands the space as the needle tip enters the submucosa. This method is less desirable with an expensive agent, because it frequently leads to spilling into the lumen or injection into the peritoneal cavity. For expensive agents, the submucosa can be identified by

inserting the needle into a thin portion of the lesion or adjacent mucosa and lifting the needle toward the lumen, so that the shape of the needle in the submucosa is visible. Small amounts of fluid are then injected until a small bleb develops, followed by additional injection. In all cases, locating the submucosa is facilitated by approaching the lesion tangentially as opposed to an en face approach.

Injection should employ a strategy designed to make snare resection easier. For EMR, 3 general strategies are followed. For moderately sized flat lesions (10–30 mm) that are entirely visible, the best approach is usually central injection through the lesion, so that the lesion becomes centered over a dome of submucosal fluid. Injection through the lesion is only contraindicated if features suggesting cancer are present. In case of sessile lesions or when a dominant nodule is present, injection should start on the most accessible side of the lesion. For lesions larger than 30 to 40 mm, a portion of the lesion is raised by submucosal injection and then snare resected, followed by systematic repeating of this process from one side of the lesion to the other. In most cases, resection proceeds from the rectal side to the cecal side to allow for clear visualization during the resection. The third strategy is used when a significant portion of the lesion is not visible because of location on the cecal side of a haustral fold. In this case, initial proximal edge (cecal edge) injection is utilized. Proximal edge injection may be facilitated by retroflexion, which is generally safe using colonoscopes in the cecum, ascending colon, transverse colon, and rectum. In the sigmoid and descending colon segments, it is safer to use an upper endoscope for retroflexion.

Dynamic injection refers to movement of the needle as injection proceeds. This usually involves lifting the colonoscope tip and needle toward the lumen center, and withdrawing the colonoscope tip and catheter distally as the submucosal mound develops. Dynamic injection allows shaping the submucosal injection mound with the goal of optimizing the ease of snare resection and directing fluid into the upper submucosa to maximize separation from the MP.<sup>13</sup>

The approach to lifting with ESD varies slightly compared with EMR. In foregut lesions, the first step is to place thermal markings 2 to 5 mm around the lesion. Once marked, it is important to start the injection outside the thermal markings, as this is where the incision will start. Injecting into the lesion is avoided during ESD to reduce bleeding in the submucosa, resulting in an obscure dissection plane, and to minimize the risk of seeding, if cancer is present. Once the steps of incision and “trimming” of the muscularis mucosa have been completed, then special attention to direct the dynamic lift into the submucosal space is performed prior to dissection. Precision during this step limits injection into vessels and the MP. Most Western endoscopists use VSs during ESD; however, NS-based solutions can be used with or without the augmentation of a water-jet/pump system.

**Table 1.** Lifting Agents

| Agent or adjuvant                 | Description/components                                                                                            | Cushion duration   | Considerations for use                                                                               | Benefits                                                                                                                                                                                                                        | Disadvantages                                                                                                                                                    |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------|--------------------|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NS                                | 0.9% saline in water                                                                                              | Shorter            | <ul style="list-style-type: none"> <li>- EMR</li> <li>- POEM</li> <li>- ESD: dissection</li> </ul>   | <ul style="list-style-type: none"> <li>- Lower price</li> <li>- Widely available</li> </ul>                                                                                                                                     | <ul style="list-style-type: none"> <li>- Quick dissipation of submucosal lift</li> </ul>                                                                         |
| Contrast agent                    | Methylene blue or indigo carmine                                                                                  | –                  | Adjuvant for: <ul style="list-style-type: none"> <li>- EMR</li> <li>- ESD</li> <li>- POEM</li> </ul> | <ul style="list-style-type: none"> <li>- Facilitates visualization of lesion border and identification of potential muscular injury</li> </ul>                                                                                  |                                                                                                                                                                  |
| Epinephrine                       | Diluted concentration of 1:50,000 - 1:500,000                                                                     | –                  | Adjuvant for: <ul style="list-style-type: none"> <li>- EMR (cold and hot)</li> </ul>                 | <ul style="list-style-type: none"> <li>- Limits the risk of immediate/acute bleeding.</li> <li>- Facilitates a clear resection field</li> </ul>                                                                                 | <ul style="list-style-type: none"> <li>- Depending on concentration used, anatomic location and quantity, may result in post-procedure pain.</li> </ul>          |
| Viscous agents                    |                                                                                                                   |                    |                                                                                                      |                                                                                                                                                                                                                                 |                                                                                                                                                                  |
| Eleview (Medtronic)               | Water, sodium chloride poloxamer 188, polyoxyl-15-hydroxystearate, medium-chain triglycerides, and methylene blue | Longer             | <ul style="list-style-type: none"> <li>- EMR</li> <li>- ESD</li> </ul>                               | <ul style="list-style-type: none"> <li>- Sustained lift</li> <li>- Non-toxic</li> <li>- Pre-packaged (5- and 10-mL vials) for convenience</li> </ul>                                                                            | <ul style="list-style-type: none"> <li>- Expensive</li> <li>- Bubble formation, which may interfere with ESD visibility during dissection</li> </ul>             |
| EndoClot SIS (Olympus)            | Absorbable modified polymers derived from plant starch particles                                                  | Longer             | <ul style="list-style-type: none"> <li>- EMR</li> <li>- ESD</li> </ul>                               | <ul style="list-style-type: none"> <li>- Potential hemostatic effects</li> <li>- Sustained lift</li> </ul>                                                                                                                      | <ul style="list-style-type: none"> <li>- Expensive</li> <li>- Requires a contrast agent addition and compounding in room.</li> </ul>                             |
| EverLift (Laborie)                | Water, glycerin, hydroxyethyl cellulose, benzyl alcohol, sodium and potassium phosphate, methylene blue           | Longer             | <ul style="list-style-type: none"> <li>- EMR</li> <li>- ESD</li> </ul>                               | <ul style="list-style-type: none"> <li>- Pre-packaged vials</li> <li>- Sustained lift</li> </ul>                                                                                                                                | <ul style="list-style-type: none"> <li>- Expensive</li> </ul>                                                                                                    |
| Blue Boost (Micro-Tech Endoscopy) | Premixed sodium hyaluronate with methylene blue                                                                   | Longer             | <ul style="list-style-type: none"> <li>- EMR</li> <li>- ESD</li> </ul>                               | <ul style="list-style-type: none"> <li>- Pre-packaged vials (10 mL)</li> <li>- Sustained lift</li> </ul>                                                                                                                        | <ul style="list-style-type: none"> <li>- Expensive</li> <li>- 50 mL max use per patient</li> </ul>                                                               |
| LiftUp (Ovesco)                   | Water, poloxamers, sodium chloride, methylene blue                                                                | Longer             | <ul style="list-style-type: none"> <li>- EMR</li> <li>- ESD</li> </ul>                               | <ul style="list-style-type: none"> <li>- Pre-packaged vials (20 mL)</li> <li>- Can be used with an inflation device</li> <li>- Sustained lift</li> <li>- Widely available</li> <li>- Nontoxic</li> <li>- Inexpensive</li> </ul> | <ul style="list-style-type: none"> <li>- Expensive</li> <li>- Easier injection if refrigerated.</li> <li>- Can turn into a gel with warm temperatures</li> </ul> |
| SG                                | Colloidal solution with similar oncotic pressure to albumin                                                       | Longer than saline | <ul style="list-style-type: none"> <li>- EMR</li> </ul>                                              | <ul style="list-style-type: none"> <li>- Sustained lift</li> <li>- Widely available</li> <li>- Nontoxic</li> <li>- Inexpensive</li> </ul>                                                                                       | <ul style="list-style-type: none"> <li>- Contraindicated in patients with gelatin hypersensitivity</li> </ul>                                                    |
| HES                               | Colloidal volume expanding as a 6% solution                                                                       | Longer than saline | <ul style="list-style-type: none"> <li>- EMR</li> </ul>                                              | <ul style="list-style-type: none"> <li>- Inexpensive</li> <li>- Longer lift with less volume compared with NS for polyps <math>\geq 30</math> mm</li> <li>- Nontoxic</li> </ul>                                                 | <ul style="list-style-type: none"> <li>- Requires a contrast agent addition and compounding in room.</li> </ul>                                                  |

Table 1. Continued

| Agent or adjuvant | Description/components                            | Cushion duration | Considerations for use | Benefits | Disadvantages                                                                                                                        |
|-------------------|---------------------------------------------------|------------------|------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------|
| HA                | Glycosaminoglycan with high viscosity             | -                | -                      | -        | - Limited due to risk of growth of residual tumor cells                                                                              |
| HPMC              | Cellulose derivative with viscoelastic properties | -                | -                      | -        | - Limited use due to risk of antigen-antibody reactions                                                                              |
| FM                |                                                   | -                | -                      | -        | - Limited use due to risk of viral contamination and transmission                                                                    |
| D50               | Hypertonic solution                               | -                | -                      | -        | - Limited use due to risk of painful inflammatory response similar to post-coagulation syndrome if injected into pericolic mesentery |

D50, 50% dextrose; EMR, endoscopic mucosal resection; ESD, endoscopic submucosal dissection; FM, fibrinogen mixture; HA, hyaluronic acid; HES, hydroxyethyl starch; HPMC, hydroxypropyl methylcellulose; NS, normal saline; POEM, peroral endoscopic myotomy; SG, succinylated gelatin.

For POEM, or any other submucosal tunneling technique, a dynamic lift is used to create a robust submucosal bleb to facilitate seamless entry into the submucosa. Whether performing gastric or esophageal POEM, once the submucosal space is entered, the subsequent injections will be directly into the submucosa. This step allows for a clear and safe submucosal dissection plane until the myotomy is performed.

Normal Saline With or Without Adjuvant Agents

NS, with or without epinephrine, has been the foremost agent used for a submucosal lift during therapeutic endoscopic interventions. The benefits of NS include low cost, easy preparation, lack of tissue damage, and widespread availability (Table 1).<sup>14</sup> Some practitioners like the “soft” cushion created by saline compared with that created by some VSs. A softer cushion can be more easily indented by the snare during EMR, particularly cold EMR, ensuring adequate depth of resection through the submucosa. The main deterrent to NS is rapid dissipation, leading to an abbreviated lift. A meta-analysis found NS resulted in fewer en bloc resections and higher rates of residual lesion compared with VSs for colorectal EMR, with no difference in adverse events. For colonic EMR, the lift duration was shortest with NS plus epinephrine (less than 3 minutes) and longest with hyaluronic acid (HA) (22 minutes).<sup>15</sup>

Contrast agents provide clear border delineation and easy identification of muscle injury. Contrast stains the submucosa blue, but the MP remains white. Thick parallel white bands indicate type 3 MP injury, which requires reinforcement with clips to prevent delayed perforation (Figure 1).

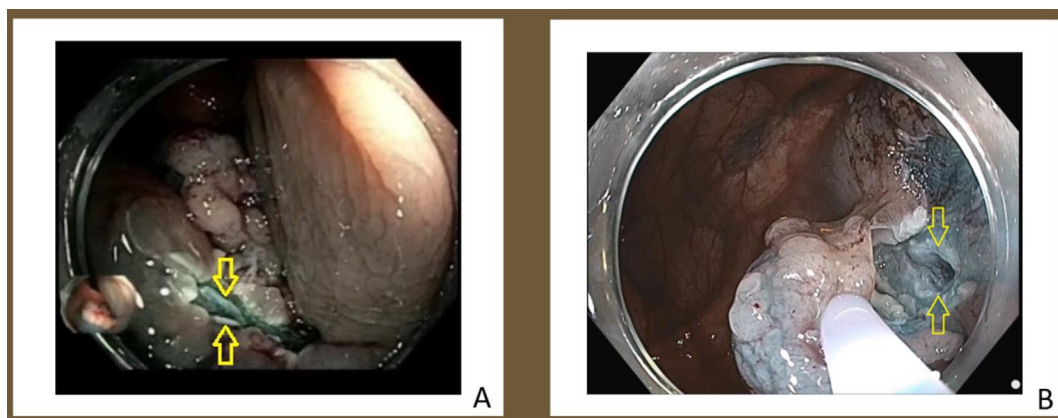
Clinicians should consider NS with a contrast agent when performing polypectomy of intermediate size NPCRLs or gastric polyps (10–19 mm) and POEM. Clinicians should avoid the use of NS with a contrast medium as the only agent used when performing ESD in the gastrointestinal tract. Due to the rapid dissipation of NS-based agents, it is important to utilize a viscous agent during ESD, especially when performing the initial lift and incision.

Viscous Lifting Agents

VSs used for EMR and ESD lead to higher rates of complete resection,<sup>16</sup> higher rates of en bloc resection (Figure 2), lower rates of residual polyp, and no differences in adverse events.<sup>17</sup>

One of the earliest VSs was 50% dextrose (D50). D50 fell out of use because it caused a painful inflammatory response similar to post-coagulation syndrome if inadvertently injected into the pericolic mesentery.<sup>18</sup>

Succinylated gelatin (SG) was the first VS investigated in a randomized controlled trial (RCT). SG was superior



**Figure 1.** (A and B) Thick parallel white bands (arrows) indicate Type 3 MP injury.

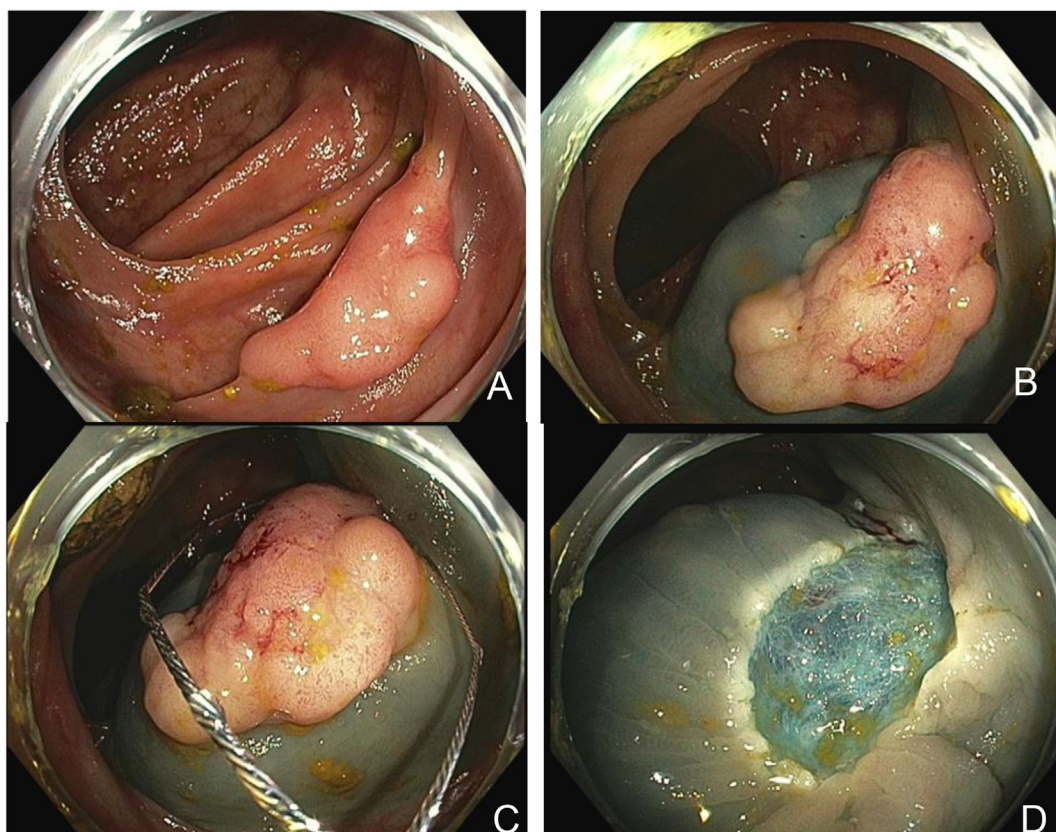
to NS, requiring less volume to maintain adequate lift, fewer injections, and permitting complete removal in fewer pieces.<sup>19</sup> SG is not available in the United States.

Hydroxyethyl starch (HES) is a colloidal volume-expanding solution previously used to treat hypovolemia. It is relatively inexpensive, and an RCT for lesions  $\geq 30$  mm in size found HES, compared with NS, required less volume and produced a longer lasting lift with lower total procedure time for EMR.<sup>15</sup>

Other VSs include glycerol, HA, hydroxypropyl methylcellulose (HPMC), and fibrinogen mixture (FM).

These agents are not routinely used in Western settings due to risks such as tissue damage at higher concentrations, antigen-antibody reactions (HPMC), viral contamination and associated transmission (FM), and growth of residual tumor cells (HA). Glycerol has limited availability in the United States.<sup>20</sup>

SIC-8000 (Eleview) became the first FDA-approved agent in 2015. SIC-8000 comes in 5- and 10-mL containers and is composed of water, sodium chloride poloxamer 188 (bulking and structural agent), polyoxyl-15-hydroxystearate (emulsifier), medium-chain-triglycerides,



**Figure 2.** (A) Non-granular laterally spreading tumor (LST) in the colon with a sessile component (Paris 0-IIA+1s). (B) Lifted LST. (C) Snare going around the LST. (D) Post-EMR base with blue stained submucosa.

and methylene blue.<sup>14</sup> In a randomized double blind trial comparing NS with SIC-8000 for NPCRLs  $\geq 20$  mm, SIC-8000 outperformed NS with regard to total volume needed to complete EMR and volume needed for initial lift. There were no differences in number of reinjections, time to resect, total procedure time, or adverse events.<sup>21</sup> Subsequently, a post-marketing investigator-initiated trial found SIC-8000 superior to hetastarch (HES).<sup>22</sup> Anecdotally, SIC-8000 in ESD results in bubble formation in the resection field, possibly related to the emulsifier, reducing visibility and dissection speed. The FDA approved SIC-8000 as a Class II medical device rather than a drug. This designation simplified FDA approval for new solutions, and the next FDA approval was for ORISE (Boston Scientific). However, ORISE led to distorted post-resection scars that were misinterpreted by some colonoscopists as submucosal tumor, and Boston Scientific removed ORISE from the world market in December 2022.<sup>23</sup>

EverLift was FDA-approved in 2020. EverLift is composed of water, glycerin, hydroxyethyl cellulose, benzyl alcohol, sodium, potassium phosphate, and methylene blue.<sup>24</sup> The solution comes in pre-made 5- and 10-mL syringes. Like other VSs, EverLift is hypertonic to facilitate sustained lifting.

EndoClot submucosal injection solution (SIS) (Olympus Corporation) is composed of absorbable modified polymers derived from plant starch particles. Ex vivo and in vivo studies found EndoClot SIS produced a longer-lasting elevation with clearer margins compared with Eleview and NS.<sup>25</sup> A potential benefit for EndoClot SIS is hemostatic effect, possibly limiting immediate bleeding during EMR and ESD. However, EndoClot SIS requires in-room compounding, which can lengthen procedure times.

Blue Boost (Micro-Tech) and LiftUp received FDA approval in 2021 and 2022. Both agents are pre-mixed with methylene blue. The active ingredient in Blue Boost is sodium hyaluronate, whereas the active ingredient in LiftUp is poloxamers, which are in a thermo-sensitive gel formulation. Both agents provide prolonged lifting.

## Device-assisted Lifting

Devices have been developed to enhance lifting. The ERBEJET 2 provides an adjustable high-pressure water jet to facilitate lifting. The ERBEJET 2 is commonly used for POEM and can be used during submucosal tunneling and endoscopic resection and ESD. The ERBEJET 2 used with a HybridKnife utilizes hydrosurgery to create lift and tissue separation. This device may be most beneficial in third-space endoscopy, especially in the setting of submucosal fibrosis.<sup>26</sup>

## Conclusion

Multiple submucosal injection solutions are available to endoscopists. An ideal lifting solution would be readily

available, inexpensive, pre-mixed, and provide a sustained submucosal cushion. However, this ideal solution currently does not exist. Injection fluids should, therefore, be selected based on planned resection method, predicted histology, local expertise and preferences, and cost.

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#### Conflicts of interest

These authors disclose the following: Tobias Zuchelli is a consultant for Boston Scientific Corporation. Amit Patel is a consultant for Medpace, Renexxion, and Sanofi. Douglas K. Rex is a consultant for Olympus Corp, Boston Scientific Inc, Braintree Laboratories, Norgine, Acacia Pharmaceuticals, Laborie Medical, Sebel Pharmaceuticals Inc, Medtronic, Exact Sciences, and ERBE; received research support from Olympus Corporation, Medivators, and ERBE USA Inc; and has ownership interest in Satisfai Health. The remaining author discloses no conflicts. The contents of this manuscript do not represent the views of the Department of Veterans Affairs or the United States Government.